

1. What Are You Building?

Give your project a working title and a 2–3 sentence summary. Who is it for, what problem does it solve, and why do you care about building it?

Amazon Review Engagement Project

This project will use data from January 2023 to June 2023 in 4 merchandise categories obtained from the [UCSD Amazon Reviews dataset](#). The SI 511 portion of this project will use Claude Code to clean data for the construction of a bipartite graph, and a category network done in SI 507. Once the SI 507 project is completed, there will be an interactive web visualization created with Claude Code for SI 511.

Stakeholder Objectives

- Business / Product
 - Reveal insights about product categories with user review data to guide resource allocation into specific tech-related categories (see appendix)
- Self
 - Learn to integrate class concepts in an un-siloed manner
 - 564: SQL & Databases
 - 507: Python Programming II
 - 511: Vibe Coding
 - Develop in-depth data architecture skills
 - Develop domain knowledge (product / business analytics) for target career
 - Improve SQL, Python, and Claude Code proficiency

2. Core Features

List your minimum viable features and any stretch goals. Each core feature should get one sentence explaining what it does from the user's perspective. Your project should hit at least four of these complexity markers (You can also propose your own):

Features	Description
Overview dashboard	Displays by category: • Retention rate • Entering users • Expansion-out rate • Expansion-in rate • Sankey diagram (reviewers → selected categories)
Calculation tooltips / side panel with assumption clarification	E.g. metric logic
TBD	TBD

- Multiple pages or components with shared state
- Persistent data (database, file storage, or API-backed)
- External API integration
- Non-trivial logic or data processing
- Authentication or role-based access
- Real-time features
- Responsive, thoughtful UX

3. Timeline

A simple week-by-week plan with milestones: prototype, core features done, testing and polish, final demo prep. This should take you to the last day of the class.

Week 1 March 23 - March 29

- SI 564 project handoff
- Data cleaning

- Cleaning data and joining datasets ~6M rows

Week 2 March 30 - April 5

- SI 507 project handoff & integration

Week 3 April 6 - April 12

- Integration with graph data
 - Visualizations, etc.

Week 4 April 13 - April 19

- Prototype website
 - Features:
 - TBD

Tips

- "I'll build a to-do app" won't pass. "I'll build a collaborative study planner with spaced repetition, calendar sync, and group analytics" will.
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- Depth beats breadth. Three deeply integrated features are better than ten shallow ones.
- If you're unsure about something, say so. That's good planning, not weakness.
- If you're not sure your idea is complex enough or whether a tool is allowed, ask before you start.

Think big. The AI can help you build more than you expect — your job is to aim high and steer well.

Appendix

Questions this project answers

- Which product categories are the strongest **entry points** for attracting first-time user engagement?
- Which categories are most effective at **retaining** users through repeated engagement over time?
- Which entry categories are most likely to drive **downstream expansion** into other high-retention categories?
- Which categories function as **bridges** that connect users to broader category exploration?
- Which categories attract high initial engagement but show weak retention or limited downstream expansion?
- Which categories create the greatest potential value for **acquisition, retention, and cross-category growth**?

Question type	Example question	Business decision supported
Entry	Which categories attract first-time users?	Where to focus discovery and acquisition efforts
Retention	Which categories sustain repeat engagement?	Where to invest in lifecycle and retention strategy

Expansion	Which categories lead users into other categories?	How to sequence recommendations and cross-sell
Portfolio	Which categories combine entry + retention + expansion strength?	Which categories deserve strategic prioritization